

Data-Driven Statistical Analysis of Supplier Lead Time Variability and Operational Risk Factors in Supply Chains

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**Data-Driven Statistical Analysis of Supplier Lead Time Variability and Operational Risk Factors in
Supply Chains**

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Abstract

Supply chain delivery performance is a critical determinant of operational efficiency and customer satisfaction yet lead time variability remains one of the most consequential and least systematically addressed sources of uncertainty in supply chain operations. This study analyzes 177,659 cleaned order records from the DataCo Global Supply Chain Dataset to identify and quantify the operational factors most strongly associated with lead time variability and delivery delays. Following data preparation in Microsoft Excel, statistical analysis was conducted in Minitab using descriptive statistics, one-way ANOVA, Tukey's post-hoc comparison, Levene's test for equal variances, the Kruskal-Wallis non-parametric test, multiple regression, and an I-MR control chart. Results indicate that shipping mode is the dominant driver of lead time variability, accounting for 34.50% of total variation ($F = 31,164.38$, $p < 0.001$). An interesting finding was that First Class shipping carried a late delivery rate of 100%, which is not due to logistics failure but due to a systematic mismatch between scheduled and actual delivery windows, with every order promised in one day while the carrier consistently delivered in two. Standard Class, despite the longest average lead time of 3.996 days, had the lowest late delivery rate at 39.76%, confirming that scheduling accuracy is a stronger predictor of on-time performance than delivery speed. The geographic region was statistically significant but practically negligible, as all 22 regions fell within a 0.119-day range. The process is out of statistical control due to structural separation between shipping modes rather than random variation, according to the I-MR control chart. These findings provide specific, quantified guidance for supply chain managers seeking to improve delivery reliability and supplier performance evaluation.

Chapter1: Introduction

1.1 Background

Supply chain management has evolved from a back-end logistics function into a central strategic concern for organizations operating across global markets. The ability to deliver products constantly and on schedule has become a defining differentiator in the market, particularly in the era of global e-commerce, and customer expectations for delivery speed and consistency have increased. Delivery lead time is defined as the time that has passed between order placement and order fulfillment, among the major performance dimensions that supply chain managers keep a watch on, which stands out as both highly visible to customers and difficult to consistently regulate. If lead times are reliable, companies can plan inventory levels more accurately, hold less safety stock, and keep replenishment cycles tight without locking up too much working capital. But when lead times are inconsistent, problems quickly follow, including stockouts, slipped production schedules, expensive delays, and a gradual erosion of customer trust (Mohammed & Mandal, 2024).

During the COVID -19 pandemic, disruptions in the global chain brought renewed urgency to the issue of lead time management, highlighting its critical role in maintaining supply chain reliability. As lead times become unpredictable within the whole supply chain network, those organizations that used to operate efficiently in stable times found it difficult to adapt to the situation. Port congestion, limited carrier capacity, labor shortages, and fluctuating demand revealed how fragile supply chains become when they rely solely on average lead times while ignoring variability. These challenges revealed that developing resilient supply chains would need an increased comprehension and control over lead-time variability rather than just focusing on the average lead-

time (Baryannis, Validi, Dani, & Antoniou, 2019). As a result this period led to increased interest in data-driven methods for supply chain analysis which can help to identify structural sources of variability and support powerful operational planning

While delays themselves are not the primary concern since delays are a necessary component of any system operating under uncertainty, the main issue is that most organizations struggle to identify the factors that contribute to delays. Variables such as shipping mode, route location, order quantity, product type, and carrier's efficiency all influence delivery time. However, in the absence of statistical analysis, organizations tend to rely on averages, make unrealistic delivery commitments, and focus improvement efforts on factors that may have little actual impact (Bandy, Satir, & Shanker, 2016).

In extremely data-intensive situations, where large amounts of transactional data frequently go unused, this issue becomes very crucial. The growing availability of detailed supply chain datasets containing order-level information such as timestamps, routing details, carrier data, and delivery outcomes creates an opportunity to shift from intuition-based decisions to data-driven planning. By applying statistical analysis, managers can better distinguish between factors that genuinely influence delivery performance and those that only appear significant but have limited practical impact (Wang, Gunasekaran, Ngai, & Papadopoulos, 2016). Building on this approach, the present study analyzes a large publicly available supply chain dataset using multiple statistical methods.

1.2 Problem Statement

Organizations rarely use systematic statistical analysis to determine which operational factors drive lead time variability or whether measured delivery failures reflect actual logistics underperformance or inaccurate delivery promises, even though transactional supply chain data is

widely available. This distinction matters because the two causes require fundamentally different interventions; one demands operational change, while the other requires only an administrative correction, yet without formal analysis, they are treated identically, misdirecting improvement resources and leaving performance problems unresolved. The absence of a rigorous, multi-method empirical framework for decomposing lead time variability by operational factor leaves supply chain managers unable to prioritize improvement investments according to their likely impact. This study addresses that gap by applying a structured sequence of statistical analyses to 177,659 order records to identify which factors explain the most variability, distinguish statistically significant from practically meaningful differences, and produce specific, quantified recommendations for supply chain planning and carrier management.

1.3 Objectives

Three primary objectives are addressed in this study. The first objective is to characterize lead time variability across the full cleaned dataset using descriptive statistics, distributional analysis, and the coefficient of variation by shipping mode, establishing a baseline understanding of the scale and structure of the dataset's variability before formal hypothesis testing.

The second objective is to determine whether shipping mode, order region, or other operational factors are statistically significant predictors of lead time and late delivery rates, using a multi-method analytical framework that includes one-way Analysis of Variance (ANOVA), Tukey's post-hoc comparison test, Levene's test for equal variances, the Kruskal-Wallis non-parametric test, multiple regression analysis, and Statistical Process Control (SPC) charting.

The third objective is to interpret the practical meaning of these statistical findings in the context of supply chain planning and supplier performance evaluation, producing specific, quantified recommendations for operations managers based on evidence derived directly from the data rather than from general principles alone.

1.4 Scope

This study uses 177,659 completed order records extracted from the DataCo Global Supply Chain Dataset (Constante, Silva, & Pereira, 2019), retained after removing non-fulfilled transactions, including canceled, pending, and suspected fraud orders. Lead time is operationalized as the "Days for Shipping (Real)" variable, representing the actual number of days elapsed between order placement and shipment. The primary predictors examined are Shipping mode, four tiers (First Class, Same Day, Second Class, and Standard Class), and order region over 22 geographic regions are the main factors that were looked at. All data preparation was performed in Microsoft Excel, and all formal statistical analysis was conducted in Minitab Statistical Software. The analysis does not extend to product category, order quantity, seasonal demand patterns, or carrier-specific performance characteristics, each of which is identified as a direction for future research in Chapter 6.

Chapter 2: Literature Review

2.1 Lead Time Variability in Supply Chains

Lead time variability, or the duration of the period it takes for orders to be fulfilled after they are placed, is one of the most challenging aspects of managing a supply chain. It influences inventory planning, production scheduling, and customer service operations. Organizations must keep more safety stock than they need when they aren't sure how long it will take for an order to arrive. This will raise costs and tie up working capital, which could be used for other things. Mohammed and Mandal (2024) looked over this issue within both manufacturing and retail supply chains, using a mix of simulation modeling and empirical data, and found that businesses with a high lead time variability consistently exhibited higher levels of stockouts, surplus inventory accumulation, and diminished order fulfillment efficiency. Their work also pointed out that companies running lean or just-in-time operations were hit hardest by this variability, since those systems have almost no buffer to absorb unexpected delays. The present study takes this understanding as its starting point and moves further by attempting to answer a question Mohammed and Mandal leave open in their simulation setting which specific operational factor, when analyzed using real transactional data at scale, accounts for the largest measurable portion of that variability.

Not all lead time variability is equally damaging, and this distinction is important for how organizations decide where to focus their improvement efforts. Bandaly, Satir, and Shanker (2016) tackled this question directly through simulation-based optimization in a multi-echelon supply chain environment. Their central finding was that only high levels of lead time variability, not moderate or low levels, justify the cost and effort of implementing coordinated, system-wide risk management responses.

This is a practically important result because it tells managers not to overreact to every fluctuation they see in delivery data. For the present study, this finding shapes a key methodological decision: because the dataset used here contains 177,659 order records, even very small differences between groups will register as statistically significant. The analysis was therefore designed from the outset to evaluate not just whether differences are statistically detectable but whether they are large enough to matter operationally. This distinction becomes most visible in the regional analysis presented in Section 4.11, where a statistically significant predictor turns out to explain less than three hours of lead time difference across twenty-two global regions.

Beyond the direct costs of holding extra inventory, lead time variability creates a broader kind of instability that travels upstream through the supply chain. Chatfield, Kim, Harrison, and Hayya (2004) showed through simulation experiments that lead time variability is one of the driving forces behind the bullwhip effect, the well-documented phenomenon where small fluctuations in customer demand get amplified into much larger swings in orders and inventory as they move upstream through a supply chain from retailer to distributor to manufacturer. Their work makes a particularly important point: it is the variability in lead time, not the average length of it, that does the most damage to supply chain stability. A supply chain with a longer but predictable lead time will generally perform better than one with a shorter but erratic lead time, because predictability allows planning systems to work accurately. This principle directly shaped the analytical approach taken in the present study. Rather than simply reporting average lead times by shipping mode, the analysis explicitly examines variability through the standard deviation and the coefficient of variation for each group, and the histogram in Section 4.2 shows exactly the kind of discrete, mode-driven clustering that Chatfield and colleagues describe—a sharp spike at two days for First Class shipments and a broad plateau across three to six days for the other modes.

2.2 Operational Risk Factors and Delivery Performance

Understanding which specific operational conditions contribute most to delivery delays has been a consistent focus in the supply chain performance literature. Among these factors, selection of the transport means, and delivery window fulfillment were identified as important variables that could be adjusted by companies without having to reconstruct their logistics systems altogether. One of the important contributions to the development of supply chain management performance measurement approaches was made by Gunasekaran, Patel, and Tirtiroglu (2001), who suggested that on-time delivery ratio was a crucial performance indicator, which measured the overall efficiency of logistics and transportation processes. What is particularly useful about their framework for the present study is the emphasis on delivery performance as a composite outcome, not the result of any one factor, but the product of multiple interacting operational variables. This is exactly the analytical challenge the present study takes on: decomposing the overall late delivery rate by shipping mode to determine which modes are driving underperformance and for what reasons.

Tao, Liang, and Bushuev (2023) took this thinking further by examining the supply chain delivery process as a system of interconnected activities, each of which contributes to whether a customer receives their order on time. A key argument in their work is that meeting the delivery window that was promised to the customer is the defining goal of supply chain logistics not simply minimizing transit time. Faster is not better if the delivery promise was set too aggressively relative to what the carrier can reliably achieve. This distinction between actual delivery performance and the accuracy of the delivery promise set at the time of order turned out to be directly relevant to the findings of the present study. The First-Class shipping analysis in Section 4.10 shows that every single one of the 27,814 First Class orders in the dataset was delivered in exactly two days perfectly consistent performance yet every one of those orders was recorded as late because the

scheduled delivery window was set at one day. The problem identified by Tao, Liang, and Bushuev (2023), a mismatch between what is promised and what is achievable, is playing out in this dataset in its most extreme possible form.

Bushuev (2018) examined this mismatch problem from a more mathematical angle, looking at how organizations could improve their on-time delivery rates across supply chains with different types of delivery time distributions. His discovery was clear and useful: if a carrier always delivers within a certain time frame, the organization can greatly improve its measured on-time rate by simply changing its delivery promise to match that window, without making any changes to how it works. This isn't a new way to plan logistics; it's a way to fix the calibration. The First Class finding in Section 4.10 of the present study is as close to a textbook illustration of Bushuev's (2018) argument as one could find in real data. A carrier delivering every order in exactly two days, promised in one day, generates a 100% late rate that disappears the moment the scheduled window is corrected to reflect reality. No new carriers, no new routes, no operational investment required.

The idea that operational pressures and performance differences vary significantly across different groups, whether industry sectors, production scales, or logistics categories, has been demonstrated empirically in several contexts. Xu, Mathiyazhagan, Govindan, Haq, Ramachandran, and Ashokkumar (2013) conducted a comparative study of how Indian manufacturing industries across four sectors responded to Green Supply Chain Management pressures, using one-way ANOVA and Tukey's post hoc test to identify which sectors differed significantly and by how much. Their study found that even within the same national regulatory environment, different operational groups produced meaningfully different performance profiles. The methodological approach in Xu et al. (2013) is directly mirrored in the present study, ANOVA followed by Tukey's

pairwise comparison is applied to shipping modes rather than industrial sectors, and the same logic applies: the goal is not just to confirm that differences exist but to identify exactly which pairs of groups are driving those differences. Baryannis, Validi, Dani, and Antoniou (2019) reinforced this point from a risk management perspective, noting in their extensive review of supply chain risk literature that one of the most consistently identified gaps in practice is the failure of organizations to use the operational data they already collect to systematically diagnose where delivery under-performance is coming from. The present study is a direct response to that gap, applying formal statistical methods to a large operational dataset to identify the root causes of late deliveries rather than simply reporting that they occur.

2.3 Data-Driven Approaches in Supply Chain Analysis

Over the past ten years, the use of statistical methods to look at large amounts of supply chain data has grown a lot. This is because businesses have more transactional records and computers have made it easier to do advanced analysis. Wang, Gunasekaran, Ngai, and Papadopoulos (2016) conducted a comprehensive survey of this domain, analyzing the application of big data analytics in logistics and supply chain functions. Their review established clearly that statistical analysis of large operational datasets could surface patterns and performance anomalies that would never be visible in summary reports or dashboards. They specifically identified delivery performance measurement and the identification of operational conditions associated with elevated risk as areas where data-driven analysis produces actionable value both of which are central objectives of the present study. The dataset used here, with 177,659 individual order records spanning four shipping modes and twenty-two geographic regions, is exactly the kind of operational data source Wang and colleagues describe as capable of generating insights unavailable through traditional aggregate reporting.

The application of ANOVA and regression to supply chain performance data has been validated in recent empirical work. Aigbedo (2024) used multiple regression, ANOVA, and chi-square analysis on fifteen years of secondary supply chain performance data to examine relationships between supply chain effectiveness and organizational outcomes. This research validates the effectiveness of such techniques for the analysis of performance differences among operational groups based on observations and not through experiments. It is critical to highlight that one of the significant methodologies identified by Aigbedo that is relevant to this study is the fact that extremely large samples make hypothesis testing able to identify statistically significant differences that are operationally insignificant; thus, effect sizes should be reported along with p-values. This caution is applied explicitly in Section 5.4 of the present study, where all twenty-two order region coefficients are statistically significant in the regression model but the total range across all regions is shown to be just 0.119 days, a difference that is real in a statistical sense but meaningless for any supply chain planning decision.

Chatterjee and Chatterjee (2024) reviewed the development of business analytics applications in supply chain management over the past decade, documenting a clear shift from basic descriptive reporting toward diagnostic and predictive analysis. Their systematic review found that the analytics applications producing the greatest operational impact were those directly tied to specific management decisions, carrier selection, lead time planning, delivery window setting, rather than those generating statistical findings without clear managerial interpretation. This focus on decision-relevant analysis shapes the interpretive approach throughout the present study. Every statistical finding is evaluated not just for its significance, but for what it implies for a supply chain manager making a concrete operational decision, whether that is choosing a shipping mode, setting

a delivery promise, or deciding where to invest improvement resources. As Chatterjee and Chatterjee (2024) emphasize, the value of supply chain analytics lies in changing decisions, not in producing p-values, and the First-Class scheduling mismatch finding in this study meets that standard directly it identifies a specific, correctable problem with a quantifiable impact of 27,814 orders currently misclassified as late.

In cases where assumptions required for performing parametric tests may be questionable, conducting both a parametric test and its non-parametric counterpart is an excellent way of coming to strong conclusions. The Kruskal-Wallis test was invented by Kruskal and Wallis back in 1952 as a non-parametric method for conducting analysis of variance when conditions of normality and homogeneity of variances could not be met. If the results from such a test match those from the corresponding parametric test, the results are said to be more reliable because they are distribution-free. This is the case with ANOVA and Kruskal-Wallis test results obtained in the current research. This robustness-checking approach is particularly important here because Levene's test confirms the equal variance assumption is violated across shipping modes, making the Kruskal-Wallis result in Section 4.7 an essential confirmation rather than a redundant one.

Statistical Process Control provides the framework for distinguishing common cause, the natural, random background variation inherent to any process, from special cause variation that is identifiable and correctable. The I-MR chart is established as the appropriate tool for monitoring individual continuous measurements over time, with common cause variation requiring system-level redesign and special cause variation addressable through targeted intervention once the source is identified. Scordaki and Psarakis (2005) demonstrated the applicability of SPC control charts in non-manufacturing commercial environments including logistics and service operations, establishing that these methods are not limited to production line contexts. Their work provides

the direct methodological justification for applying the I-MR chart to supply chain delivery data in Section 4.9 of the present study, where the out-of-control signals confirmed the presence of special cause variation attributable to the structural separation between shipping modes a finding that motivated the mode-specific analysis and recommendations developed in Chapters 5 and 6.

The DataCo Global Supply Chain Dataset used throughout this study was compiled and published by Constante, Silva, and Pereira (2019) to support exactly this kind of supply chain analytics research, providing individual order-level records with both actual and scheduled delivery times that make the comparative analysis of on-time performance across operational groups possible.

2.4 Summary

The literature reviewed in this chapter builds a clear and connected argument that runs directly into the present study. Lead time variability is a well-researched operational issue whose consequences extend from direct inventory costs to systemic supply chain instability, and whose root causes in any specific supply chain cannot be assumed but must be investigated empirically (Mohammed & Mandal, 2024; Chatfield et al., 2004). The research consistently identifies carrier selection and delivery window accuracy as among the most controllable and impactful drivers of on-time delivery performance, and it specifically highlights the gap between what is promised and what is consistently achieved as a source of measured underperformance that is administrative rather than operational in nature (Tao, Liang, & Bushuev, 2023; Bushuev, 2018).

The methodological framework built around ANOVA, Tukey's post-hoc comparison, Levene's test, Kruskal-Wallis, multiple regression, and SPC control charting is grounded in validated applications across supply chain and logistics research contexts, with the important requirement that statistical significance always be accompanied by effect size interpretation to distinguish

findings that should drive decisions from those that are operationally negligible (Aigbedo, 2024; Kruskal & Wallis, 1952; Scordaki & Psarakis, 2005; Xu et al., 2013). What the literature does not provide is a study that applies all these methods together to a large operational transactional dataset to simultaneously characterize lead time variability, identify its primary driver, and distinguish statistically significant from practically meaningful differences across both shipping modes and geographic regions. The present study fills that gap using 177,659 order records from the DataCo Global Supply Chain Dataset (Constante, Silva, & Pereira, 2019), producing findings that are statistically rigorous, practically interpretable, and directly actionable for supply chain managers.

Chapter 3: Methodology

3.1 Data Source

The DataCo Global Supply Chain Dataset was selected for this analysis. It was compiled and published by Constante, Silva, and Pereira (2019) and made publicly available through Mendeley Data. The dataset is widely used in supply chain analytics research due to its breadth of operational variables and its coverage of a realistic multi-regional distribution network. It contains transactional records capturing order placement dates, shipping dates, actual delivery days, scheduled delivery days, shipping modes, product categories, order quantities, delivery statuses, and regional identifiers spanning 22 geographic regions. The dataset's in detailed structure with individual order-level records that include both scheduled and actual delivery times makes it particularly well-suited for lead time variability analysis, as it allows not only the calculation of actual lead times but also the direct comparison of actual versus scheduled performance, enabling the construction of a binary delay indicator that forms the basis of the late delivery rate calculations in Section 4.10.

3.2 Data Preparation

The raw dataset contained 180,520 records. Prior to processing analysis, records were filtered in Microsoft Excel to retain only orders representing completed delivery transactions. Orders classified as Cancelled, Pending, Pending Payment, On Hold, Suspected Fraud, and Processing were not included. Because they do not reflect completed deliveries and hence do not offer reliable observations of real lead time, these categories were left out. After filtering, 177,659 records remained, representing a retention rate of 98.4% of non-excluded records. Lead time was operationalized using the "Days for Shipping (Real)" variable, which records the actual number of days

elapsed between order placement and shipment for each individual transaction. A binary indicator variable, "Delay," was constructed to flag orders where actual shipping days exceeded scheduled shipping days (Delay = 1 for late, Delay = 0 for on-time or early). This variable was used subsequently to calculate late delivery rate for each shipping mode that was calculated in Excel using COUNTIFS functions to determine the proportion of orders where the Delay variable equaled 1 relative to the total orders in that mode by shipping mode and is the basis for Table 5 in Section 4.10.

3.3 Exploratory Analysis

Exploratory data analysis was conducted before formal statistical testing to identify initial patterns and verify data integrity. Excel pivot tables were created to calculate the average lead time by order region and shipment mode producing a summary of mean lead time across all 22 geographic regions. These summaries provided an initial characterization of variation across operational groups and guided the selection of variables for formal statistical modeling. The late delivery rate calculation revealed a striking 100% rate for First Class shipments during the exploratory phase, which motivated the detailed investigation of scheduled versus actual delivery windows that became a central finding of the study. The distribution of lead times was also examined visually through a histogram generated in Minitab to assess the shape of the distribution and identify structural clustering patterns.

3.4 Statistical Analysis

All formal statistical analysis was conducted in Minitab Statistical Software following a sequential structure in which each analysis informed and justified the next. Nine analyses were performed in the following order.

First, overall descriptive statistics were calculated for the lead time variable across all 177,659 records, including the mean, standard deviation, variance, coefficient of variation, median, quartiles, minimum, and maximum. The coefficient of variation was included to enable scale-independent comparison of variability across groups with different means a metric particularly relevant when comparing shipping modes with substantially different average lead times.

Second, a histogram was generated to examine the shape of the lead time distribution and assess whether it followed a normal pattern or exhibited structural clustering that would indicate systematic operational behavior.

Third, to allow for a direct group-level comparison of central tendency and variability, the descriptive statistics were computed independently by shipping mode, paying special emphasis to the coefficient of variation as a gauge of relative unpredictability for each mode.

Fourth, A box plot was then created using lead time on the Y-axis and mode of shipping as the categorical variable to compare central tendency, dispersion, and shape of the distribution for all four modes.

Fifth, a one-way Analysis of Variance (ANOVA) was performed, considering lead time as the dependent variable and shipping method as the independent variable. In this instance, the null hypothesis stated that the means of the four alternative shipping methods did not differ in terms of lead times. Significance level was chosen to be $\alpha = 0.05$. Tukey's post-hoc pairwise comparison test was conducted immediately following the significant ANOVA result to identify which specific pairs of shipping modes differed significantly from each other, using simultaneous 95% confidence intervals for all pairwise mean differences.

Sixth, Levene's test for equality of variances was conducted to assess whether the homoscedasticity assumption underlying the ANOVA was satisfied across the four shipping mode

groups. This test was performed to determine whether a non-parametric robustness check was warranted.

Seventh, the Kruskal-Wallis non-parametric test was conducted as a robustness check given the violation of equal variance and normality assumptions confirmed by Levene's test and the histogram. This test uses rank-based methods and does not assume equal variances or normally distributed data, making it the appropriate non-parametric alternative when distributional assumptions cannot be confirmed (Kruskal & Wallis, 1952)

Eighth, a multiple regression model was estimated with lead time as the response variable and both shipping mode and order region entered simultaneously as categorical predictors. Both variables were entered in the same model to assess their independent contributions to lead time while controlling for each other. Four-in-one residual diagnostic plots were generated to assess model fit and confirm the pattern of residuals was consistent with the discrete, bounded nature of the lead time variable.

Ninth, an Individuals and Moving Range (I-MR) control chart was constructed on the full lead time series to assess whether the delivery process operates in a state of statistical control or exhibits special cause variation attributable to identifiable operational factors such as shipping mode.

3.5 Tools

Microsoft Excel was used for data cleaning, filtering, lead time calculation, delay variable construction, and exploratory pivot table analysis.

Minitab Statistical Software was used for all formal statistical analyses including descriptive statistics, histogram generation, boxplot construction, one-way ANOVA with Tukey's comparison, Levene's test for equal variances, Kruskal-Wallis non-parametric test, multiple regression with residual diagnostics, and I-MR control chart generation.

Chapter 4: Results

4.1 Descriptive Statistics

The first step in the analysis was to get a clear overall picture of lead time across the entire cleaned dataset before breaking it down by any grouping variable. Descriptive statistics were calculated for the Days for Shipping (Real) variable across all 177,659 order records.

Overall, the mean lead time across all orders was 3.554 days with a standard deviation of 1.574 days. The coefficient variation of lead times was 44.30%, indicating that variability in this dataset is nearly half the size of the mean which itself a high level of relative variability for a logistics process. The median sat at 3 days, Q1 at 2 days and Q3 at 5 days, giving an interquartile range of 3 days. Lead times went anywhere from 0 days on the low end all the way up to 6 days. The variance was 2.479 days squared, and the standard error came out to 0.00374 days, which is quite small and the full output is shown in Figure 1 with a summary provided in Table 1.

Statistics												
Variable	N	N*	Mean	SE Mean	StDev	Variance	CoeffVar	Minimum	Q1	Median	Q3	Maximum
Days for shipping (real)	177659	0	3.55396	0.003735	1.57442	2.4788	44.3	0	2	3	5	6

Figure 1. Descriptive Statistics for Days for Shipping (Real) - Minitab Output

Table 1. Summary of Descriptive Statistics for Lead Time

Statistics	Value
N (observations)	177,659
Mean	3.554 days
Standard Error of Mean	0.00374
Standard Deviation	1.574 days
Variance	2.479

Coefficient of Variation	44.30%
Minimum	0 days
First Quartile (Q1)	2 days
Median	3 days
Third Quartile (Q3)	5 days
Maximum	6 days

4.2 Lead Time Distribution

To analyze the form of the lead time distribution across all 177,659 order records, a histogram was generated in Minitab.

The histogram displays two different features instead of a smooth bell-shaped distribution centered on the mean. The first is a very pronounced spike at exactly 2 days far taller than any other bar in the chart. The second is a relatively flat plateau running from 3 days through 6 days, with each of those bars at roughly similar heights. There is very little recorded at 0 or 1 day. This non-normal, bimodal-like shape immediately indicates that the spike at 2 days is triggered by a specific subgroup of orders rather than random fluctuations. Every First-Class order in the dataset was delivered in precisely 2 days with zero variance, as shown in Section 4.3, and it was those 27,814 orders that caused the spike. Second Class and Standard Class orders with significantly longer and more uniformly distributed lead times can be seen by the plateau, which ranges of 3 to 6 days. Figure 2 shows the result.

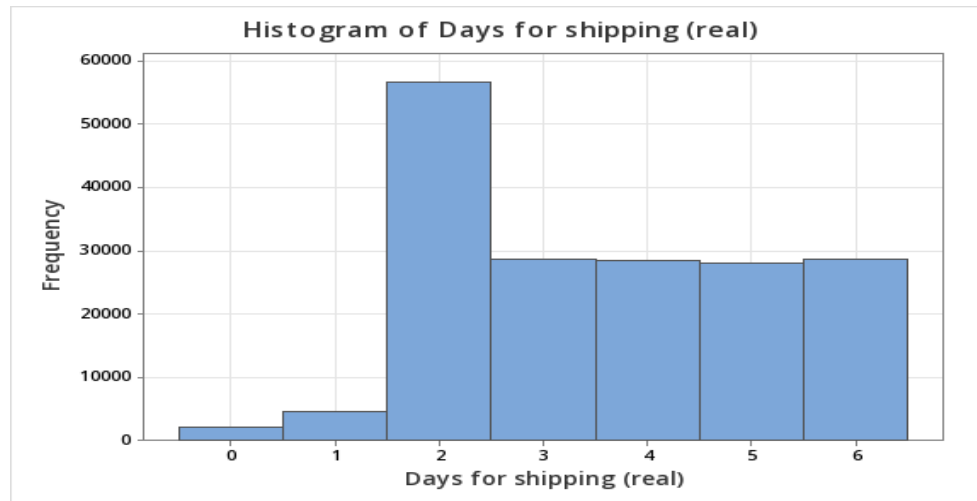


Figure 2. Histogram of Days for Shipping (Real)

4.3 Lead Time by Shipping Mode

Descriptive statistics were calculated separately for each of the four shipping modes to enable direct comparison of central tendency and variability across groups.

The variations between modes are remarkable. First Class had a mean lead time of exactly 2.000 days with a standard deviation of 0.000 and a coefficient of variation of 0%. This means every single one of the 27,814 First Class orders in the dataset arrived in exactly 2 days not approximately 2 days, but precisely 2 days without a single exception. From a process consistency standpoint, this is about as reliable as a delivery system can get.

Same Day had the lowest mean lead time at 0.677 days, which makes sense given what Same Day shipping is designed to do. However, it also had the highest coefficient of variation at 69.05%, meaning that while orders tend to arrive quickly, there is considerable inconsistency in exactly how quickly. A CoefVar of 69% is very high and means Same Day lead times are genuinely unpredictable relative to their own average.

Second Class and Standard Class told a very similar story to each other. Second Class had a mean of 3.991 days and a CoefVar of 35.48%, while Standard Class had a mean of 3.996 days and a CoefVar of 35.47%. These two modes are nearly identical in both their average lead time and their variability profile, despite being presented as different service tiers. The full output is shown in Table 2 summarizes the key metrics for each mode.

Table 2. Descriptive Statistics and Coefficient of Variation by Shipping Mode

Ship- ping Mode	N	Mean (days)	StDev	Coef- Var (%)
First Class	27,814	2.000	0.000	0.00
Same Day	6,877	0.677	0.468	69.05
Second Class	35,216	3.991	1.416	35.48
Stand- ard Class	107,752	3.996	1.417	35.47

The boxplot in Figure 3 visualizes these differences clearly. First Class shows up as a single horizontal line at 2 days reflecting zero variance. Same Day has a small compact box near the bottom of the chart. Second Class and Standard Class both show wide boxes with long whiskers stretching from roughly 2 days up to 6 days, and their boxes overlap almost completely, which visually confirms what the descriptive statistics already suggested these two modes are delivering essentially the same lead time experience.

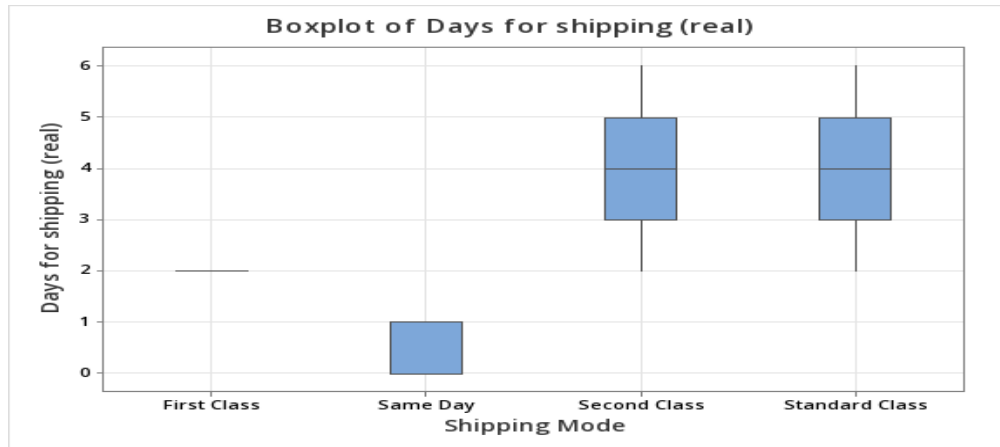


Figure 3. Boxplot of Lead Time by Shipping Mode

4.4 ANOVA Results

The descriptive comparison in Section 4.3 showed clear differences in average lead time across shipping modes, but summary statistics alone cannot confirm whether those differences are statistically significant. A one-way Analysis of Variance was conducted to formally test whether the mean lead times across the four shipping modes were equal.

The null hypothesis was that there was no significant difference between the means of the four groups. From Table 3, it can be observed that the F-value obtained from the ANOVA is 31,164.38 and the p-value is 0.000. Therefore, the null hypothesis is rejected. The R-squared value of 34.50% indicates that shipping mode alone explains more than one-third of all lead time variability observed across the 177,659 orders, confirming it as the dominant driver of lead time in this dataset. The residual plots are shown in Figure 8B.

Table 3. One-Way ANOVA Results for Lead Time by Shipping Mode

Source	DF	F-Value	P-Value	R-Squared
Shipping Mode	3	31,164.38	0.000	34.50%
Error	177,655	-	-	-

Total	177,658	-	-	-
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4.5 Tukey Post-Hoc Comparison

A significant ANOVA result confirms that at least one pair of shipping modes has different mean lead times but does not identify which specific pairs are responsible. Tukey's post-hoc comparison test was conducted to answer that question, comparing all six possible pairs of shipping modes simultaneously while maintaining an overall 95% confidence level across all comparisons.

Standard Class and Second Class both received the letter A, meaning their mean lead times of 3.996 and 3.991 days are not statistically distinguishable from each other despite being marketed as different service tiers. Every other pair of modes is statistically significantly different, with First Class receiving the letter B and the same day receiving the letter C.

The Tukey simultaneous 95% confidence intervals for each paired mean difference are displayed in Figure 4. Any interval not crossing the zero line confirms a statistically significant difference between that pair. The only interval crossing zero is the Standard Class versus Second Class comparison, consistent with both modes sharing grouping letter A.

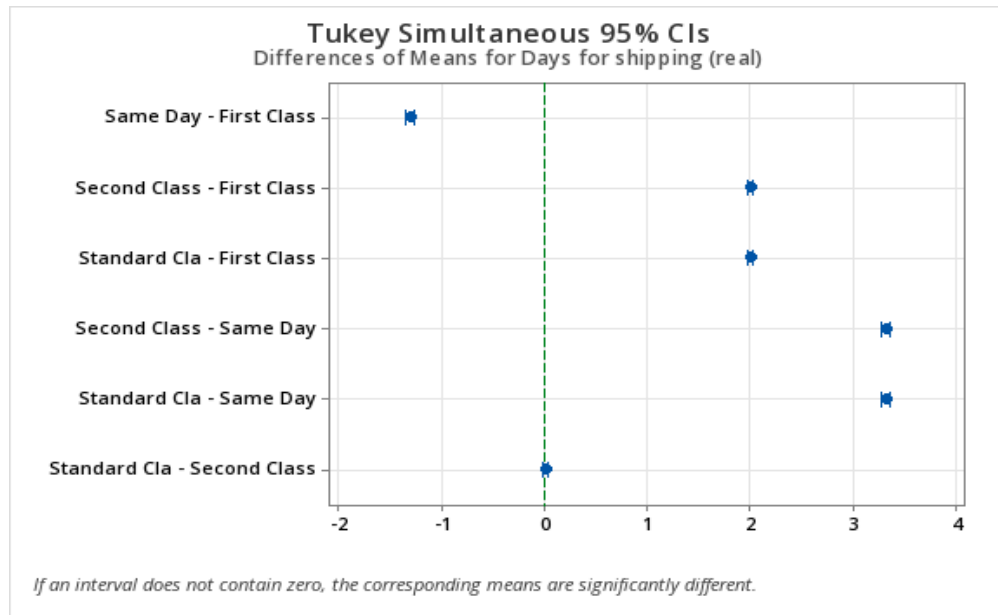


Figure 4. Tukey Simultaneous 95% Confidence Intervals - Differences of Means by Shipping Mode

Figure 5 presents the interval plot showing the mean lead time and 95% confidence interval for each mode individually. Same Day sits near 0.677 days, First Class at exactly 2.000 days, and Second Class and Standard Class nearly overlap near 4.000 days making the separation between groups visually clear and confirming the Tukey grouping results numerically established in Table 4.

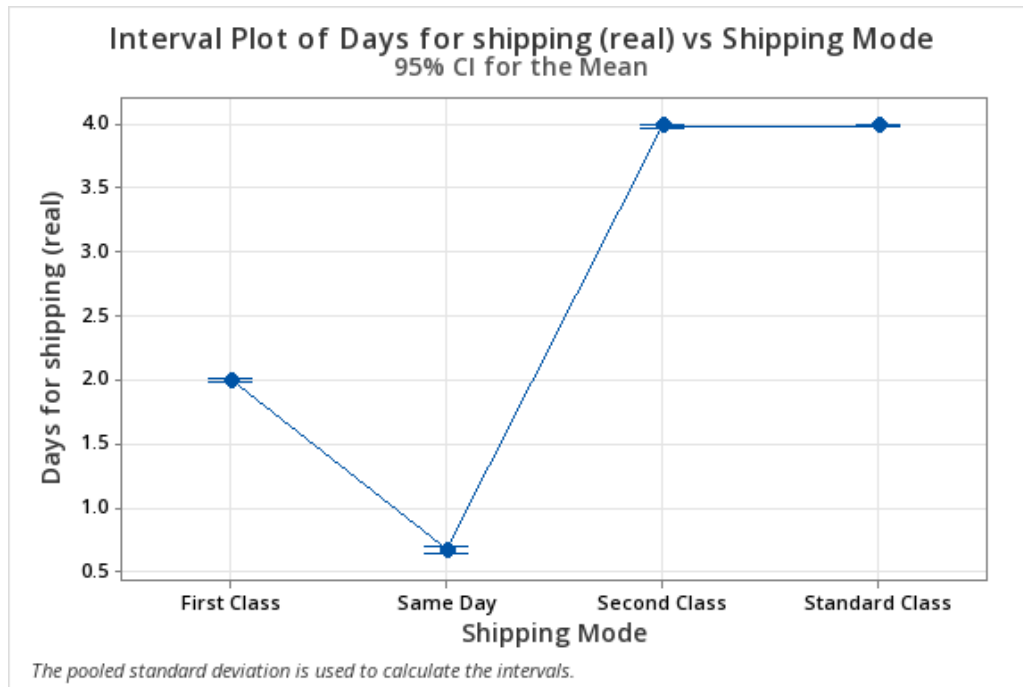


Figure 5. Interval Plot of Lead Time by Shipping Mode (95% CI for the Mean)

Table 4. Tukey Post-Hoc Grouping Results

Shipping Mode	N	Mean (days)	StDev	Tukey Group
Standard Class	107,752	3.996	1.417	A
Second Class	35,216	3.991	1.416	A
First Class	27,814	2.000	0.000	B
Same Day	6,877	0.677	0.468	C

4.6 Levene's Test for Equal Variances

One of the assumptions underlying one-way ANOVA is that the variance in the response variable is roughly equal across all groups being compared a property called homoscedasticity. Given what the descriptive statistics already showed First Class with zero variance and Same Day with a coefficient of variation more than double that of Second and Standard Class there was reason to suspect this assumption might not hold. Levene's test was conducted to formally check it.

The findings, displayed in Figure 6, validated that concern. With a p-value of 0.000 and a Levene's test statistic of 4,663.42, the variances among the four shipping modalities aren't equal. Bonferroni 95% confidence intervals for each mode's standard deviation in the same figure do not overlap which indicates that the variance differences are statistically significant and real. First Class has a standard deviation of zero, Same Day has a standard deviation of 0.468, and both Second Class and Standard Class have standard deviations around 1.416. Because the equal variance assumption is violated, the ANOVA results should be interpreted with some caution. This is why the Kruskal-Wallis test was included as a follow-up to confirm whether the ANOVA conclusion holds when no distributional assumptions are made at all.

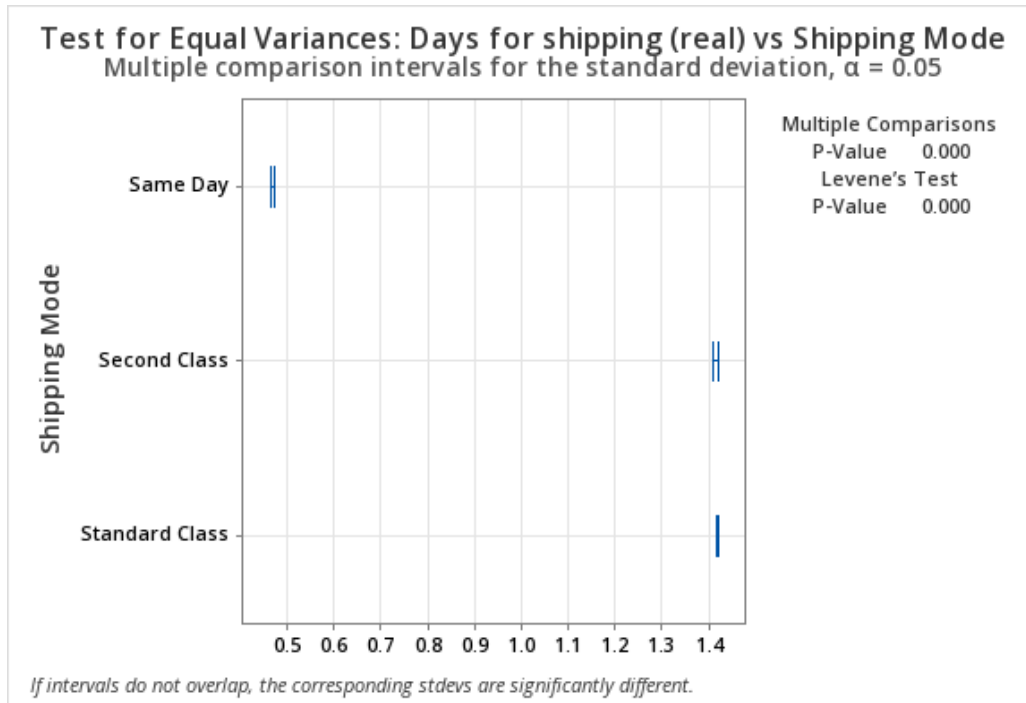


Figure 6. Levene's Test for Equal Variances by Shipping Mode - Minitab Output

4.7 Kruskal-Wallis Non-Parametric Test

An alternate method of performing the one-way ANOVA test is via the Kruskal-Wallis test, which uses ranked data instead of actual figures without any assumption on the shape of distribution and equal variance among groups.

The value of H test considering the ties is 64,132.92, and its corresponding p-value is 0.000; therefore, it is proven that there are significant differences between the lead time distributions of the delivery services. The median lead times are: Same Day = 1 day, First Class = 2 days, Second Class = 4 days, Standard Class = 4 days. The mean rank for Standard Class was 103,415 the highest of any mode and Same Day had the lowest mean rank at 3,439. The conclusion is consistent with the ANOVA result shipping modes differ significantly in their lead time distributions, and this holds without relying on any parametric assumptions. The results are shown in Figure 7.

Kruskal-Wallis Test: Days for shipping (real) versus Shipping Mode					
Descriptive Statistics					
Shipping Mode	N	Median	Mean Ra	Z-Value	
First Class	27814	2	35186.5	-190	
Same Day	6877	1	3439	-141	
Second Class	35216	4	103247	58.91	
Standard Class	107752	4	103415	148.8	
Overall	177659		88830		
Test					
Null hypothesis	H ₀ : All medians are equal				
Alternative hypothesis	H ₁ : At least one median is different				
Method	DF	H-Value	P-Value		
Not adjusted for ties	3	60992	0		
Adjusted for ties	3	64133	0		

Figure 7. Kruskal-Wallis Test Results by Shipping Mode - Minitab Output

4.8 Regression Analysis

While the ANOVA and Kruskal-Wallis tests examined shipping mode alone, a multiple regression model was estimated to assess whether geographic region also contributed independently to lead time differences. Both shipping mode and order region were entered simultaneously as categorical predictors so that each variable's contribution could be evaluated while controlling for the other.

The model summary in Figure 8a shows an R-squared of 34.50% and an adjusted R-squared of 34.49%. Adding 22 regional categories to the model barely moved the R-squared beyond what shipping mode alone already explained, which is the first indication that region contributes very little once shipping mode is accounted for.

The coefficients in Figure 8A show that Same Day orders arrive 1.323 days sooner than First Class on average after controlling for region, while Second Class and Standard Class take approximately 2 additional days compared to First Class. All 22 regional coefficients were statistically significant, but their values ranged only from 0.128 to 0.248 days the largest regional effect

in the entire model is less than a quarter of a day. What this means operationally is discussed in Section 5.4.

Model Summary					
S	R-sq	R-sq(adj)	R-sq(pred)		
1.27433	34.50%	34.49%	34.48%		
Coefficients					
Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	1.8418	0.042	43.84	0	
Shipping Mode					
Same Day	-1.3233	0.0172	-77.07	0	1.2
Second Class	1.9908	0.0102	194.68	0	1.82
Standard Class	1.99601	0.00857	232.8	0	1.92
Order Region					
Caribbean	0.1462	0.0438	3.34	0.001	9.24
Central Africa	0.2476	0.052	4.76	0	2.73
Central America	0.1579	0.0422	3.74	0	25.81
Central Asia	0.2193	0.0691	3.18	0.001	1.56
East Africa	0.1693	0.0511	3.31	0.001	2.89
East of USA	0.1897	0.0443	4.29	0	7.9
Eastern Asia	0.175	0.0441	3.96	0	8.22
Eastern Europe	0.1565	0.0462	3.38	0.001	5
North Africa	0.1524	0.0472	3.23	0.001	4.29
Northern Europe	0.1403	0.0435	3.23	0.001	10.6
Oceania	0.1536	0.0434	3.54	0	10.89
South America	0.1585	0.0428	3.7	0	15.25
South Asia	0.1693	0.044	3.85	0	8.68
South of USA	0.1583	0.0462	3.43	0.001	5.08
Southeast Asia	0.1529	0.0435	3.51	0	10.36
Southern Africa	0.1761	0.0561	3.14	0.002	2.19
Southern Europe	0.1277	0.0435	2.93	0.003	10.27
US Center	0.1508	0.0447	3.37	0.001	6.88
West Africa	0.1605	0.0466	3.45	0.001	4.74
West Asia	0.144	0.0447	3.22	0.001	7.03
West of USA	0.1408	0.0439	3.21	0.001	8.94
Western Europe	0.1747	0.0422	4.14	0	24.83

Figure 8 A. Regression Model Summary, Coefficients, Minitab Output

The residual plots in Figure 8b show the expected non-normal, discrete pattern consistent with the distributional findings already established in Sections 4.2 and 4.6.

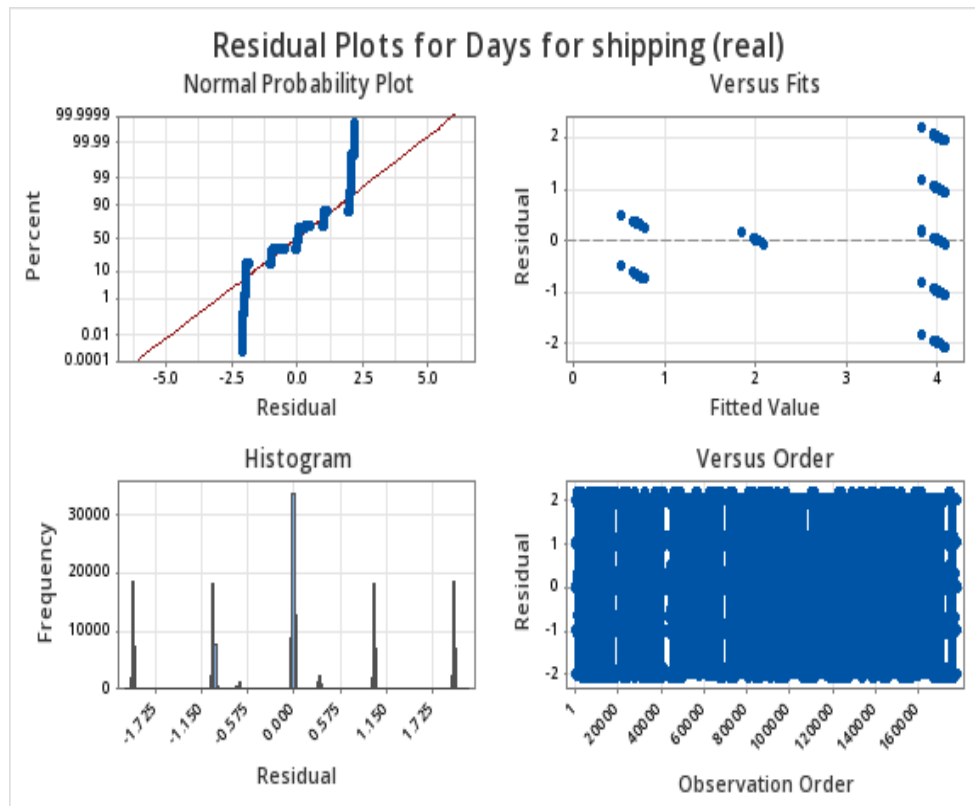


Figure 8 B. Residual Plots Minitab Output

The regression ANOVA table in Figure 8 C confirms this clearly. Shipping mode produced an adjusted sum of squares of 151,809 with an F-value of 31,161.13 and $p = 0.000$. Order region produced an adjusted sum of squares of only 74 with an F-value of 2.07 and $p = 0.002$. Both predictors are statistically significant, but shipping mode explains over 2,000 times more variance than region, the difference in practical contribution between the two predictors is substantial.

Analysis of Variance					
Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	25	151919	6076.7	3742.04	0
Shipping Mode	3	151809	50603	31161.13	0
Order Region	22	74	3.4	2.07	0.002
Error	177633	288461	1.6		
Lack-of-Fit	66	183	2.8	1.71	0
Pure Error	177567	288278	1.6		
Total	177658	440379			

Figure 8 C. Analysis of Variance from Regression

4.9 I-MR Control Chart

An Individuals and Moving Range control chart was constructed on the full lead time series to assess whether the delivery process operates in a state of statistical control.

The Individual Value chart has a center line at 3.554 days, an upper control limit of 7.177 days, and a lower control limit of -0.070 days. The Moving Range chart has a center line of 1.362 days and an upper control limit of 4.451 days. The chart flagged numerous Test 1 violations points more than three standard deviations from the center line at multiple observation points throughout the dataset. This confirms that the lead time process is not in statistical control and that the variation observed is not purely random. The interpretation of what is driving these out-of-control signals is addressed in Section 5.5. The control chart is shown in Figure 9.

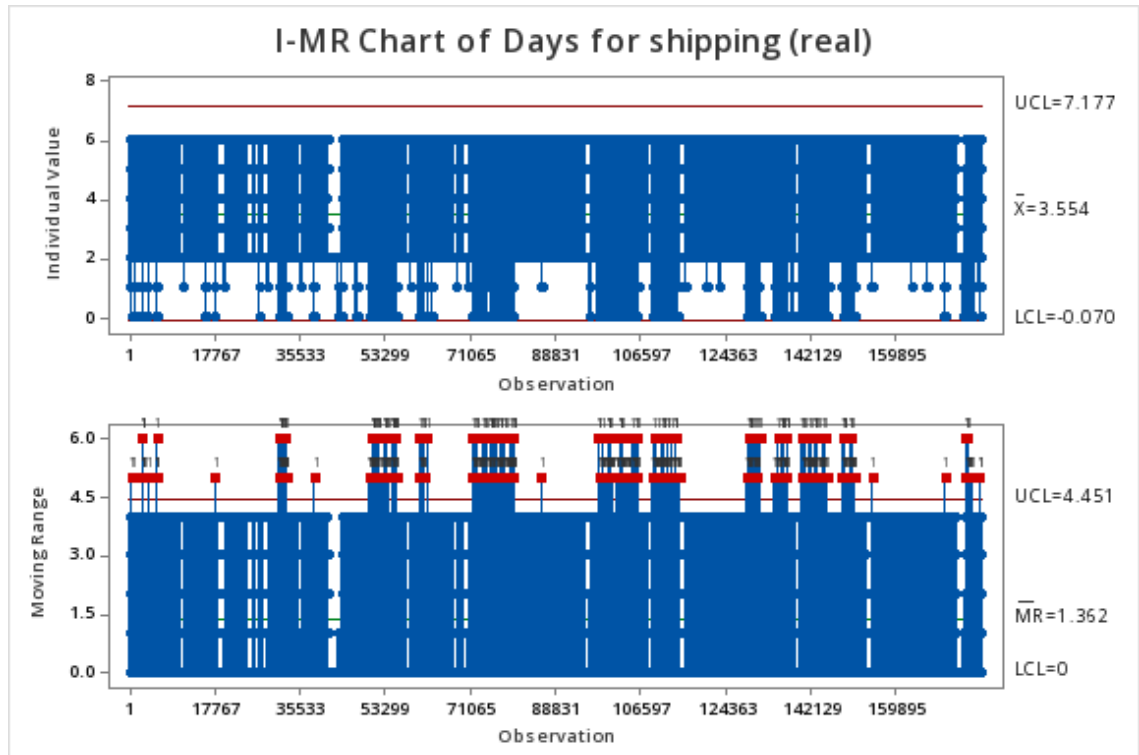


Figure 9. I-MR Control Chart of Days for Shipping (Real) -Minitab Output

4.10 Late Delivery Rate by Shipping Mode

Late delivery rate was calculated for each shipping mode using the Delay variable constructed during data preparation, which flagged any order where the actual shipping days exceeded the scheduled shipping days.

The numbers produced one of the most striking findings in the entire analysis. First Class had a late delivery rate of 100% every single one of its 27,814 orders was recorded as late. Second Class had a late delivery rate of 80.24%, Same Day had 68.26%, and Standard Class had the lowest rate at 39.76%.

On the surface this looks like a clear ranking from worst to best performance, with the fastest shipping mode performing worst and the slowest performing best. That conclusion would be wrong, and understanding why it is wrong is one of the most practically important insights this

study produces. The late delivery rate is calculated by comparing actual delivery days to scheduled delivery days it is a measure of whether reality matched the promise, not a measure of how fast orders arrived. An order can arrive quickly and still be recorded as late if the promise was set even quicker. That is exactly what is happening with First Class, and it is addressed fully in the Discussion chapter. Table 5 presents the results.

Table 5. Late Delivery Rate by Shipping Mode

Shipping Mode	N	Mean Lead Time (days)	Late Delivery Rate (%)
First Class	27,814	2.000	100.00%
Second Class	35,216	3.991	80.24%
Same Day	6,877	0.677	68.26%
Standard Class	107,752	3.996	39.76%

4.11 Regional Analysis

Average lead time was calculated for all 22 order regions. A representative selection of regional results is shown in Table 6, with the complete range spanning from 3.491 days in Southern Europe to 3.610 days in Central Africa across all 22 regions

Lead time in all 22 regions varied between 3.491 days in Southern Europe and 3.610 days in Central Africa, for a total span of 0.119 days. All regions were tightly grouped around the mean lead time in the entire data set, which was 3.554 days, with none deviating substantially from that figure. The regression results in Section 4.8 confirmed that region is a statistically significant predictor, but the practical magnitude of the regional differences equivalent to approximately 2 hours and 51 minutes across 22 global regions is addressed in the Discussion in Section 5.4.

Table 6. Average Lead Time by Order Region (Selected Regions)

Region	Avg Lead Time (days)	Region	Avg Lead Time (days)
Southern Europe	3.491	Eastern Asia	3.560
Southern Africa	3.502	West Africa	3.601
US Center	3.530	Central Africa	3.610
Eastern Europe	3.540	Western Europe	3.554
South America	3.550	Central America	3.558

Chapter 5: Discussion

5.1 Shipping Mode as the Dominant Factor

The one-way ANOVA results pointed clearly to shipping mode as the main factor driving differences in lead time across this dataset. The F-statistic came out to 31,164.38 with a p-value of 0.000, tested across 3 and 177,655 degrees of freedom, so there really is no question about whether these differences are statistically real they are.

What this finding does is confirm something that makes practical sense too. Shipping mode is not just a statistically significant variable; it matters for how supply chain planning should be approached. The R-squared value of 34.50% carries the more important practical message: a single categorical operational variable, the choice of shipping mode accounts for more than one-third of all lead time variability observed across nearly 178,000 individual orders. The remaining 65.50% of variability is attributable to factors not captured in this model, including order-specific characteristics, within-mode carrier performance variation, and random operational noise, all of which represent directions for future investigation.

This finding aligns with the broader supply chain literature. Tao, Liang, and Bushuev (2023) argue that carrier selection and shipping mode are among the most consequential structural decisions in supply chain design because they establish the baseline delivery performance characteristics around which all downstream planning inventory positioning, safety stock calculation, replenishment cycle design, and customer promise-setting must be organized. The present results provide quantitative empirical support for this argument: the choice of shipping mode does not merely shift the average delivery time but fundamentally reshapes the variability profile of the delivery system, as demonstrated by the striking differences in coefficient of variation across

modes from 0% for First Class to 69.05% for Same Day (Table 2). As Gunasekaran, Patel, and Tirtiroglu (2001) established, on-time delivery rate is a composite outcome of multiple interacting factors, and the present ANOVA and Tukey results demonstrate that shipping mode is the single most powerful of those factors in this dataset.

Using rank-based techniques that do not need equal variances or normally distributed data, the Kruskal-Wallis result of $H = 64,132.92$ with $p = 0.000$ supports the ANOVA conclusion with lower statistical assumptions. Methodologically significant, the agreement between the parametric ANOVA and the non-parametric Kruskal-Wallis test shows that the shipping mode differences are robust and not artifacts of distributional assumptions that Levene's test revealed to be broken (statistic = 4,663.42, $p = 0.000$). The Levene's test finding itself carries operational significance: because variances differ dramatically across modes from zero variance for First Class to a standard deviation of 1.416 for Standard Class, supply chain planners cannot treat lead time as a single homogeneous process. Each shipping mode requires its own safety stock calculation based on its own variance, its own control limits for process monitoring, and its own scheduling standards. The Tukey post-hoc grouping further reveals that Second Class and Standard Class, despite their different tier names and presumably different pricing, deliver statistically identical lead times. Organizations paying premium rates for Second Class over Standard Class should examine whether the service tier differentiation is reflected in actual delivery performance, as the data suggest it is not.

5.2 The First-Class Scheduling Problem

The finding that First Class shipping carries a 100% late delivery rate is the most operationally significant result of this study, and it requires careful and precise interpretation to avoid a

fundamental misdiagnosis. The raw statistic 100% late delivery rate taken in isolation would suggest catastrophic carrier failure. The descriptive statistics in Table 2 tell an entirely different story. First Class has a standard deviation of exactly 0.000 days and a coefficient of variation of 0%. Every single one of the 27,814 First Class orders in this dataset was delivered in exactly 2 days without a single exception across more than 27,000 transactions. The carrier's performance is, by any operational measure, perfectly consistent and perfectly reliable. The 100% late delivery rate exists entirely because the system recorded a scheduled delivery time of 1 day for First Class orders while the carrier consistently achieves 2 days. Every order is measured against a standard it was never operationally designed to meet.

It is important to make a distinction between a logistics challenge and a scheduling challenge as they require separate actions. In case of an underperforming carrier, the proper course of action may include performance evaluations, renegotiation of the existing contracts, and even changing carriers which is both time- and money-intensive. Because the issue is scheduling miscalibration, the entire problem is solvable through a single administrative change: updating the scheduled lead time for First Class from 1 day to 2 days. This requires no operational modification, no capital investment, no renegotiation with the carrier, and no process redesign. It would immediately reclassify all 27,814 First Class orders from "late" to "on time," reducing the system-wide measured late delivery rate by approximately 29% without a single operational change.

Bushuev (2018) formalized the theoretical basis for exactly this type of intervention, demonstrating that when actual delivery times follow a consistent and predictable statistical distribution, organizations can substantially improve measured on-time performance simply by aligning delivery window promises with the empirical distribution of actual performance. The First Class result in this dataset is a near-perfect empirical illustration of Bushuev's (2018) framework:

a delivery time distribution with zero variance and a deterministic outcome of exactly 2 days, being measured against a promised window of 1 day, produces a 100% failure rate that is entirely an artifact of the measurement standard rather than of operational reality. As Xu, Mathiyazhagan, Govindan, Haq, Ramachandran, and Ashokkumar (2013) observed in their study of operational performance differences, what appears to be a performance failure often reflects a measurement or expectation mismatch rather than an underlying operational deficiency a principle the First Class finding illustrates with clarity.

This finding also illuminates a broader principle about performance measurement systems in supply chains. Organizations frequently set delivery promises based on aspirational targets what they would like to achieve rather than empirical baselines what they consistently achieve. When aspirational and empirical performance diverge, the measured late delivery rate becomes a function of ambition rather than operations, creating misleading performance reports that misdirect management attention and improvement resources. The appropriate corrective action is to ground delivery window promises in statistical analysis of actual delivery performance, precisely the type of analysis conducted in this study.

5.3 Standard Class Scheduling Accuracy

The contrast between First Class and Standard Class performance profiles illuminates a broader and generalizable principle about what drives on-time delivery rates in supply chain operations. Standard Class had the longest average lead time at 3.996 days, a coefficient of variation of 35.47%, and a late delivery rate of 39.76% the best on-time performance of any shipping mode despite being the slowest. First Class had the shortest non-zero lead time at 2.000 days, zero variability, and a 100% late rate. This inverse relationship between speed and on-time performance is

counterintuitive only if on-time delivery is assumed to depend primarily on speed. It follows directly from the fact that on-time delivery is a relative measure: it compares actual performance to a promised standard, not to an absolute benchmark. When promised delivery windows are set conservatively allowing adequate time to accommodate natural variation in actual delivery performance a higher proportion of orders will arrive within the window even when individual lead times vary considerably.

This finding directly supports the argument developed by Chatfield, Kim, Harrison, and Hayya (2004) that reducing lead time variability not simply reducing the mean is often the more valuable supply chain intervention. The Standard Class result shows that a delivery mode with a standard deviation of 1.417 days and a CoefVar of 35.47% can still achieve a 39.76% on-time rate when its delivery windows are calibrated appropriately. An organization seeking to improve the Standard Class on-time rate should focus not on tightening the delivery promise which would increase the measured late rate without any operational improvement but on reducing the underlying variability of actual Standard Class lead times by identifying and addressing the operational factors responsible for the widespread from 2 to 6 days.

For Same Day shipping, the coefficient of variation of 69.05% presents a distinct planning challenge. Despite having the lowest mean lead time, Same Day is the most unpredictable mode for inventory and production planning purposes. An organization that schedules replenishment based on Same Day lead times faces the highest risk of unexpected shortfalls, because the actual delivery time is genuinely uncertain in a way that other modes are not. The business implication is that Same Day shipping should be reserved for situations where the exact delivery time is less critical than simply receiving the goods quickly and should not be relied upon for precision scheduling applications.

5.4 Regional Impact and Practical Significance

The regression analysis result for the order region illustrates one of the most important and frequently misunderstood aspects of statistical analysis in large-sample settings. Every one of the 22 regional coefficients was statistically significant at the 0.05 level, with p-values at or below 0.003. A surface reading of this result would suggest that geography is an important driver of lead time variability and that regional routing optimization should be a priority for supply chain improvement. The regional average lead times presented in Section 4.11 tell a fundamentally different picture.

Across all 22 order regions, average lead time ranged from 3.491 days in Southern Europe to 3.610 days in Central Africa a total spread of 0.119 days, equivalent to approximately 2 hours and 51 minutes. No supply chain planning decision safety stock levels, replenishment cycle frequency, carrier contract terms, routing strategy, or delivery window setting would be materially different based on a regional lead time difference of less than 3 hours across 22 global regions. The regional differences are statistically real with 177,659 observations; the statistical power is sufficient to detect extremely small true differences, but they are operationally meaningless.

This result supports the methodological caution emphasized by Aigbedo (2024), who demonstrated that very large datasets reliably detect differences that are real but trivial, and that effect size analysis is essential for the correct interpretation of statistical output in such settings. As Chatterjee and Chatterjee (2024) emphasize, the value of supply chain analytics lies not in generating statistically significant outputs but in producing insights that directly change operational decisions a standard that the regional finding fails to meet despite its statistical significance, and which the First-Class scheduling mismatch finding meets with direct and quantifiable impact. The conclusion for supply chain managers is clear: geographic region is not a meaningful target

for lead time improvement efforts in this context, and resources allocated to regional routing optimization based on this regression output would be misdirected. Shipping mode is where improvement resources should be focused.

5.5 Control Chart Interpretation

The I-MR control chart confirmed that the lead time process is not operating in a state of statistical control. Multiple points exceed the upper control limit throughout the observation period, which in a standard process monitoring context would signal that something unexpected is disrupting the system. In this case, however, the cause is not a random disruption, it is structural.

The out-of-control signals are being generated by the fundamental separation between shipping modes. First Class orders all land at exactly 2 days, Same Day orders cluster near 0 to 1 day, and Standard and Second Class orders spread across 3 to 6 days. When all four modes are monitored together as a single process with one center line at 3.554 days, observations from the extreme modes, particularly First Class and Same Day, consistently appear as outliers relative to that overall mean and trigger control limit violations. The chart is not detecting a process breakdown. It is detecting the fact that four structurally different delivery processes are being monitored as one.

The practical implication is straightforward. Lead time should not be monitored at the aggregate level across all shipping modes combined. Each shipping mode operates as its own distinct process with its own mean and its own natural variation, and each one needs its own control chart with its own mode-specific center line and control limits. If that were done, First Class would show a perfectly stable process at exactly 2 days with zero variation. Same Day would show a low mean with moderate variation. Second and Standard Class would show wider natural variation around a mean near 4 days. Monitoring them separately would allow genuine performance shifts within any

individual mode to be detected immediately, rather than being hidden within the noise of an aggregate chart that is structurally out of control by design.

5.6 Limitations

Several limitations constrain the interpretation of these findings. The dataset covers a single company's supply chain operations over a specific time and may not generalize to organizations in different industries, with different product mixes, or using different carrier networks. The relationships identified here, particularly the dominance of shipping mode, may differ substantially in supply chains where geographic distance, customs complexity, or product fragility are more consequential drivers of delivery variability.

The analytical framework examined shipping mode and order region as the primary predictors but did not incorporate product category, order quantity, seasonal demand patterns, week-day of order placement, or carrier-specific performance characteristics. The omission of these variables means the regression model's R-squared of 34.50% understates the total explainable variance in the data, and that some variance currently attributed to shipping mode may reflect correlations with these unmeasured variables.

The binary delay variable captures schedule adherence but does not capture all dimensions of customer-perceived delivery reliability, which may also depend on communication quality, proactive delay notification, partial shipment practices, and tracking visibility none of which are present in the DataCo dataset. The regression model assumes linear relationships with categorical predictors; interaction effects between shipping mode and region were not explored. Finally, the I-MR control chart was applied to individual observations across the full dataset rather than to mode-specific subgroups, which is the most conservative approach but one that conflates structural mode differences with genuine process instability, as discussed in Section 5.5.

Chapter 6: Conclusion

6.1 Summary of Findings

This project set out to answer a straightforward but practically important question which operational factors drive lead time variability in a supply chain, and do the factors that are statistically significant also matter in a practical sense? Starting with 177,659 cleaned order records from the DataCo Global Supply Chain Dataset (Constante, Silva, & Pereira, 2019), the analysis worked through a sequence of statistical methods in Minitab, each one building on the previous, to build a complete and evidence-based answer to that question.

The overall descriptive statistics established the baseline. The mean lead time across all orders was 3.554 days with a standard deviation of 1.574 days and a coefficient of variation of 44.30%. That coefficient of variation was the first signal that something structural rather than random was creating wide differences in delivery times across orders. A CoefVar approaching 50% in a logistics process is high, and it immediately pointed toward the need for group-level investigation rather than simply reporting the overall average.

The histogram made the structural nature of the problem visible. Instead of a smooth bell-shaped curve, the distribution showed a sharp spike at exactly 2 days and a flat plateau running from 3 to 6 days a shape consistent with carrier and mode-specific clustering rather than random distributional variation (Chatfield, Kim, Harrison, & Hayya, 2004). That observation motivated the group-level descriptive analysis, which confirmed that each shipping mode has a completely different lead time profile. First Class delivered every order in exactly 2 days with zero variance. Same Day had the lowest average at 0.677 days but the highest coefficient of variation at 69.05%, making it the most unpredictable mode despite being the fastest. Second Class and Standard Class

were nearly identical to each other in both average lead time and variability, despite being presented as different service tiers.

The one-way ANOVA formally confirmed what the descriptive analysis suggested. Shipping mode is a highly significant predictor of lead time with an F-statistic of 31,164.38 and a p-value of 0.000, explaining 34.50% of total lead time variability across the dataset. Tukey's post-hoc test identified exactly which modes differed all pairs except Second Class and Standard Class were significantly different from each other. Levene's test confirmed that the equal variance assumption was violated across modes, and the Kruskal-Wallis test confirmed the ANOVA finding without relying on any parametric assumptions, with an H-statistic of 64,132.92 and $p = 0.000$ (Kruskal & Wallis, 1952). The agreement between both tests made the conclusion about shipping mode considerably more reliable than if only one test had been used.

The regression model added order region as a second predictor alongside shipping mode. The R-squared remained at 34.50% after adding 22 regional categories, and the regression ANOVA table showed that shipping mode accounted for an adjusted sum of squares of 151,809 while order region accounted for only 74. Both predictors were statistically significant but the difference in their practical contributions was enormous. The regional coefficients ranged from just 0.128 to 0.248 days across all 22 global regions a total spread of 0.119 days, equivalent to less than 3 hours. As Aigbedo (2024) noted, large sample sizes reliably detect differences that are statistically real but operationally trivial, and this regional result is a clear example of that phenomenon.

The late delivery rate analysis produced the most striking finding of the entire project. First Class had a 100% late delivery rate every single one of its 27,814 orders was recorded as late. But the descriptive statistics showed that First Class also had a standard deviation of exactly zero,

meaning every order arrived in precisely 2 days without exception. The carrier was not underperforming.

The problem was that the scheduled delivery window was set at 1 day while the carrier consistently delivered in 2 days. This is not a logistics failure it is a scheduling miscalibration that directly illustrates the principle demonstrated by Bushuev (2018), that calibrating delivery promises to match actual carrier performance eliminates measured underperformance without any operational change. Standard Class had the lowest late delivery rate at 39.76% despite being the slowest mode, because its delivery windows were set conservatively enough to accommodate its natural variation. The I-MR control chart confirmed that the overall process is out of statistical control, with the out-of-control signals driven by the structural separation between shipping modes rather than by any genuine process breakdown (Scordaki & Psarakis, 2005).

6.2 Recommendations

Three recommendations follow directly from the findings of this project. The first and most immediately actionable is to correct the scheduled lead time for First Class shipping from 1 day to 2 days. This requires no operational change whatsoever no new carriers, no new routes, no capital investment. According to the findings, all 27,814 First Class orders were shipped exactly 2 days after the order was placed, without any variation, whereas each shipment was expected to be completed in 1 day, which was an unrealistic commitment. Correcting this single scheduling entry would immediately move all 27,814 orders from late to on-time, reducing total measured late deliveries by approximately 27% with zero operational effort.

The second recommendation is that supply chain managers should prioritize shipping mode analysis before examining any other operational factor when investigating lead time variability. Shipping mode explained 34.50% of total variability in this dataset accounting for an adjusted sum

of squares of 151,809 compared to just 74 for order region across 22 global regions. The coefficient of variation by mode is a particularly important metric that is often overlooked. Same Day shipping, despite its low average of 0.677 days, carries a CoefVar of 69.05% making it the most unpredictable mode for planning purposes. Safety stock calculations and replenishment cycles should be based on mode-specific variability profiles rather than on the overall dataset average.

The third recommendation is to monitor lead time separately by shipping mode rather than at the aggregate level. The I-MR control chart showed the overall process is out of statistical control, but this is a structural artifact of mixing four fundamentally different delivery processes into one chart. Each mode should have its own control chart with its own center line and control limits, which would make genuine performance shifts within any individual mode immediately detectable rather than hidden within aggregate noise.

6.3 Future Work

While this project addressed its core objectives, several directions remain open for future investigation that would build meaningfully on the findings established here.

The regression model explained 34.50% of total lead time variability using shipping mode and region, leaving 65.50% unaccounted for. The DataCo dataset contains additional variables including product category and order quantity that were not incorporated into the regression model in this study. Including these variables in a follow-up analysis would provide a more complete picture of what drives lead time differences and would likely increase the model's explanatory power beyond the current 34.50%. Understanding which product categories or order sizes are associated with longer or more variable lead times would also give supply chain managers more specific and targeted guidance than shipping mode analysis alone can provide.

The First-Class scheduling mismatch raises a question; this dataset cannot answer whether the problem is isolated to one carrier or reflects a broader pattern across multiple providers. If carrier level data were available, the same analysis could be run by individual carrier to determine whether the fix requires a simple scheduling update or a full contract renegotiation, which are fundamentally different interventions with very different costs and timelines.

Separate I-MR control charts for each shipping mode built using mode-specific center lines and control limits rather than pooled dataset values would give operations teams a far more sensitive monitoring tool than the aggregate chart used here. A Standard Class specific chart would detect genuine performance deterioration within that mode immediately rather than having it buried within structural noise from other modes.

Perhaps the most interesting question this project leaves open is how common the scheduling mismatch phenomenon is across supply chains more broadly. It is remarkable to note that a total of 27,814 orders delivered at a 100% late delivery rate due to mis calibrated delivery promise rather than any operational failure. This highlights the question of how much the companies might be unintentionally misrepresenting their delivery success without being aware of it. The way the supply chain industry tracks and discloses on-time delivery outcomes would be substantially altered by examining if this trend is present across various supply chain datasets and, if so, how widespread the gap between promised and achievable delivery windows tends to be.

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